

Constraint-Conditioned Proportional Switching-Time Control for a Three-Cell Switched-Affine DC–DC Converter

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Abstract

Switched-affine power converters can be regulated by changing when their operating modes switch rather than by commanding a continuous input amplitude. This paper develops a low-complexity cycle-to-cycle controller for that setting. A first-order model maps the state error at one cycle boundary and the perturbations of the nominal switching instants to the error at the next boundary. A discrete linear-quadratic design supplies a static proportional feedback gain. Because unconstrained feedback can produce infeasible switching schedules, a scalar conditioning step contracts the complete perturbation vector until every dwell interval satisfies its lower bound. The method is evaluated in simulation on a three-cell multilevel DC–DC converter with eight operating modes and a minimum dwell time of 3 μ s. Starting from a large perturbation, the normalized cycle-to-cycle error decreases from 0.9854 to 0.0086 after 80 cycles, a reduction of 99.1%, while the shortest observed dwell interval remains 3.0 μ s. For the local model, the raw-action admissible polyhedron is positively invariant, and the simulated trajectory enters it after 3 conditioned cycles. The results establish the proportional controller as a computationally inexpensive constrained baseline for periodic-orbit stabilization, while also exposing its local-model and uniform-scaling limitations.

Keywords: switched-affine systems; periodic-orbit stabilization; proportional state feedback; switching-time control; dwell-time constraints; DC–DC converter.

1 Introduction

Switched-affine systems evolve according to one of a finite number of affine vector fields. In power electronics, the active vector field is selected by the switch configuration. Practical devices cannot switch arbitrarily fast, so each mode must remain active for at least a minimum dwell time. Under such timing constraints, the meaningful steady behavior is generally a periodic orbit rather than a static equilibrium [1, 2, 3].

A useful control architecture first determines a nominal mode sequence and switching schedule that generate a desired periodic orbit, then perturbs the internal switching instants to reject cycle-to-cycle deviations. Predictive formulations can impose dwell-time constraints directly, but require a constrained optimization at each update [4, 5, 3]. This work isolates the simpler feedback law already implicit in the terminal linear-quadratic design: a static, proportional map from the cycle-to-cycle state error to switching-time perturbations.

The term *proportional controller* requires care. The proposed controller is not a conventional scalar P controller and does not manipulate voltage, current, or duty ratio directly. It is a multi-input state-feedback law whose manipulated variables are signed displacements of switching instants. Its

unconstrained output is followed by uniform dwell-time conditioning. This distinction is maintained throughout the paper and in the accompanying domain glossary.

The contribution of this draft is fourfold:

1. a concise derivation of the cycle-to-cycle proportional switching-time controller from the linearized periodic dynamics;
2. an explicit conditioning rule that guarantees the minimum dwell time without an online optimization problem;
3. a polyhedral characterization of the raw-action admissible region and its maximal positively invariant subset; and
4. a reproducible simulation study using the current MATLAB model of the three-cell converter.

2 Statement of the Problem

2.1 System Dynamics and Sequence

Consider the switched-affine plant

$$\dot{\mathbf{x}}(t) = A_{\sigma(t)}\mathbf{x}(t) + \mathbf{b}_{\sigma(t)}, \quad \mathbf{x}(t) \in \mathbb{R}^n, \quad (1)$$

where $\sigma(t)$ selects one of M operating modes. During one cycle, a fixed mode sequence $\boldsymbol{\Omega} = (\omega_1, \dots, \omega_N)$ is applied at nominal switching instants

$$0 = \bar{\tau}_0 < \bar{\tau}_1 < \dots < \bar{\tau}_N = T_p. \quad (2)$$

The associated nominal dwell intervals are $\bar{d}_i = \bar{\tau}_i - \bar{\tau}_{i-1}$.

The nominal schedule generates a target periodic orbit $\bar{\mathbf{x}}(t)$ satisfying $\bar{\mathbf{x}}(T_p) = \bar{\mathbf{x}}(0)$. Its cycle-boundary state is denoted by $\mathbf{x}^* = \bar{\mathbf{x}}(0)$. This point must not be confused with the desired average operating point about which the orbit oscillates.

For mode ω_i , introduce the augmented state and mode matrix

$$\mathcal{X}(t) = \begin{bmatrix} \mathbf{x}(t) \\ 1 \end{bmatrix}, \quad \tilde{A}_i = \begin{bmatrix} A_{\omega_i} & \mathbf{b}_{\omega_i} \\ \mathbf{0}^\top & 0 \end{bmatrix}. \quad (3)$$

Exact propagation over a dwell interval d_i is then

$$\mathcal{X}(\tau_i) = \exp(\tilde{A}_i d_i) \mathcal{X}(\tau_{i-1}). \quad (4)$$

This expression is also used by the simulation to avoid numerical integration error within each constant-mode segment.

2.2 Linearization Procedure

At cycle k , the internal switching instants are perturbed according to

$$\tau_{k,i} = \bar{\tau}_i + \delta\tau_{k,i}, \quad i = 1, \dots, N-1, \quad (5)$$

with fixed cycle boundaries $\delta\tau_{k,0} = \delta\tau_{k,N} = 0$. Consequently, the duration perturbation of interval i is

$$\delta d_{k,i} = \delta\tau_{k,i} - \delta\tau_{k,i-1}, \quad i = 1, \dots, N. \quad (6)$$

Define the cycle-to-cycle error $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}^\star$ and the nominal augmented transition of interval i as

$$\Psi_i = \exp\left(\tilde{A}_i \bar{d}_i\right). \quad (7)$$

The augmented state at the beginning of cycle k is

$$\mathcal{X}_k = \bar{\mathcal{X}}(0) + \begin{bmatrix} \mathbf{e}_k \\ 0 \end{bmatrix}. \quad (8)$$

Therefore, the exact one-cycle propagation under the perturbed durations is

$$\mathcal{X}_{k+1} = e^{\tilde{A}_N(\bar{d}_N + \delta d_{k,N})} \dots e^{\tilde{A}_1(\bar{d}_1 + \delta d_{k,1})} \left(\bar{\mathcal{X}}(0) + \begin{bmatrix} \mathbf{e}_k \\ 0 \end{bmatrix} \right). \quad (9)$$

For one interval, the matrix exponential has the first-order expansion

$$e^{\tilde{A}_i(\bar{d}_i + \delta d_{k,i})} = \Psi_i + \tilde{A}_i \Psi_i \delta d_{k,i} + \mathcal{O}(\delta d_{k,i}^2). \quad (10)$$

Substituting this expansion into (9) and retaining one perturbed factor at a time gives

$$\begin{aligned} \mathcal{X}_{k+1} &\doteq \Psi_N \dots \Psi_1 \bar{\mathcal{X}}(0) + \Psi_N \dots \Psi_1 \begin{bmatrix} \mathbf{e}_k \\ 0 \end{bmatrix} \\ &\quad + \sum_{i=1}^N \Psi_N \dots \Psi_{i+1} \tilde{A}_i \Psi_i \dots \Psi_1 \bar{\mathcal{X}}(0) \delta d_{k,i}. \end{aligned} \quad (11)$$

Here, $\doteq$ denotes equality to first order. Terms containing two timing perturbations, or a timing perturbation multiplied by $\mathbf{e}_k$, are discarded. The first factor applied to the initial state is the rightmost one, and products with no factors are the identity.

Because the nominal trajectory is periodic, $\Psi_N \dots \Psi_1 \bar{\mathcal{X}}(0) = \bar{\mathcal{X}}(0)$. Subtracting this nominal term from (11) and applying the projection $P = [I_n \ 0]$ gives [5, 3]

$$\mathbf{e}_{k+1} \doteq \Phi \mathbf{e}_k + G \delta \mathbf{d}_k, \quad G = [\mathbf{g}_1 \quad \dots \quad \mathbf{g}_N], \quad (12)$$

where

$$\Phi = e^{A_{\omega_N} \bar{d}_N} \dots e^{A_{\omega_1} \bar{d}_1}, \quad (13)$$

$$\mathbf{g}_i = P \Psi_N \dots \Psi_{i+1} \tilde{A}_i \Psi_i \dots \Psi_1 \bar{\mathcal{X}}(0). \quad (14)$$

Using (6), the duration-coordinate model reduces to the fixed-period switching-instant model

$$\mathbf{e}_{k+1} \doteq \Phi \mathbf{e}_k + \Gamma \delta \boldsymbol{\tau}_k, \quad \Gamma = [\mathbf{g}_1 - \mathbf{g}_2 \quad \dots \quad \mathbf{g}_{N-1} - \mathbf{g}_N], \quad (15)$$

where $\Phi \in \mathbb{R}^{n \times n}$ and $\Gamma \in \mathbb{R}^{n \times (N-1)}$. Equivalently, the sensitivity to internal switching instant i is

$$\Gamma_i = P \Psi_N \dots \Psi_{i+1} (\tilde{A}_i - \tilde{A}_{i+1}) \bar{\mathcal{X}}(\bar{\tau}_i), \quad i = 1, \dots, N-1. \quad (16)$$

Thus, the augmented duration linearization recovers the vector-field jump at each internal switch while preserving the fixed cycle period. Equation (15) is local: its accuracy decreases as the state and switching schedule move away from the nominal orbit.

3 Proportional Switching-Time Controller

3.1 Cost Function and Linear-Quadratic Gain

The raw control law is the static state feedback

$$\delta\tau_k^{\text{raw}} = -K e_k. \quad (17)$$

The gain is obtained from the discrete-time linear-quadratic regulator for (Φ, Γ) ,

$$K = \text{dlqr}(\Phi, \Gamma, Q, R), \quad (18)$$

which minimizes the infinite-horizon quadratic criterion associated with

$$\sum_{k=0}^{\infty} \left(e_k^\top Q e_k + \delta\tau_k^\top R \delta\tau_k \right). \quad (19)$$

Thus, *proportional* denotes a memoryless linear relation between the sampled state error and the switching-time perturbation vector. The matrix gain is synthesized from the local one-cycle model rather than tuned as an independent scalar.

3.2 Dwell-Time Constraints

The dwell interval produced by a candidate perturbation is

$$d_{k,i} = \bar{d}_i + \delta\tau_{k,i} - \delta\tau_{k,i-1}. \quad (20)$$

The hardware constraint is

$$d_{k,i} \geq d_{\min}, \quad i = 1, \dots, N. \quad (21)$$

Direct componentwise clipping of $\delta\tau_k^{\text{raw}}$ does not guarantee (21), because each internal switching instant affects two adjacent dwell intervals. Instead, define

$$\Delta d_{k,i} = \delta\tau_{k,i}^{\text{raw}} - \delta\tau_{k,i-1}^{\text{raw}}. \quad (22)$$

A single conditioning factor $\beta_k \in [0, 1]$ is selected as

$$\beta_k = \min \left\{ 1, \min_{i: \Delta d_{k,i} < 0} \frac{d_{\min} - \bar{d}_i}{\Delta d_{k,i}} \right\}, \quad (23)$$

where the inner minimum is omitted when no interval shrinks. The applied perturbation is

$$\delta\tau_k = \beta_k \delta\tau_k^{\text{raw}}. \quad (24)$$

Because all nominal intervals satisfy $\bar{d}_i \geq d_{\min}$, the ratios in (23) are nonnegative and the resulting schedule satisfies (21). The method preserves the direction of the raw LQR action, but sacrifices its magnitude whenever a dwell constraint becomes active.

The online work consists of one matrix–vector multiplication, interval differences, scalar comparisons, and one vector scaling. No quadratic program is solved. This simplicity is the main advantage over constrained predictive control, but uniform scaling can be conservative because one active interval attenuates every switching-time perturbation.

3.3 Constraint-Admissible and Invariant Regions

Let $D \in \mathbb{R}^{N \times (N-1)}$ map switching-instant offsets to dwell-duration changes,

$$\delta \mathbf{d}_k = D \delta \boldsymbol{\tau}_k, \quad D = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ -1 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 1 \\ 0 & \cdots & 0 & -1 \end{bmatrix}. \quad (25)$$

For the raw law (17), the dwell constraints are satisfied without conditioning when

$$\bar{\mathbf{d}} - DK\mathbf{e}_k \geq d_{\min} \mathbf{1}_N. \quad (26)$$

The corresponding raw-action admissible region is therefore the polyhedron

$$\mathcal{C} = \{\mathbf{e} \in \mathbb{R}^n : DK\mathbf{e} \leq \bar{\mathbf{d}} - d_{\min} \mathbf{1}_N\}. \quad (27)$$

Inside $\mathcal{C}$, $\beta_k = 1$ and the LQR action is unchanged. Outside $\mathcal{C}$, the controller is not infeasible: $\beta_k = 0$ recovers the feasible nominal schedule, and (23) selects the largest uniform scaling permitted along the raw-action direction.

Define the unconditioned local closed-loop matrix

$$A_{\text{cl}} = \Phi - \Gamma K. \quad (28)$$

The maximal positively invariant subset of $\mathcal{C}$ is

$$\mathcal{C}_\infty = \bigcap_{\ell=0}^{\infty} \left\{ \mathbf{e} \in \mathbb{R}^n : A_{\text{cl}}^\ell \mathbf{e} \in \mathcal{C} \right\}. \quad (29)$$

If A_{cl} is Schur stable, every initial error in $\mathcal{C}_\infty$ remains raw-action admissible and converges to the origin under the local linear model without activating the conditioner. This is a local linear-model certificate, not a global region-of-attraction guarantee for the nonlinear switched plant.

4 Application Examples

4.1 Multilevel Converter

The simulated plant is a three-cell multilevel DC–DC converter with $\mathbf{x} = [v_{C_1}, v_{C_2}, i_L]^\top$. For binary switch commands $u_1, u_2, u_3 \in \{0, 1\}$, its mode-dependent dynamics are

$$A_{\mathbf{u}} = \begin{bmatrix} 0 & 0 & (u_2 - u_1)/C_1 \\ 0 & 0 & (u_3 - u_2)/C_2 \\ (u_1 - u_2)/L & (u_2 - u_3)/L & -R/L \end{bmatrix}, \quad \mathbf{b}_{\mathbf{u}} = \begin{bmatrix} 0 \\ 0 \\ Eu_3/L \end{bmatrix}. \quad (30)$$

The model therefore has eight operating modes. The parameters used by the current MATLAB configuration are listed in table 1.

The MATLAB-indexed mode sequence is

$$\boldsymbol{\Omega} = (1, 2, 4, 8, 3, 1, 5, 8, 5), \quad (31)$$

Table 1: Simulation parameters from the current PATINO_2 configuration.

Quantity	Symbol	Value
Source voltage	E	30 V
Capacitances	C_1, C_2	40 μ F
Inductance	L	10 mH
Load resistance	R	10 Ω
Nominal period	T_p	286 μ s
Minimum dwell time	$d_{\min}$	3 μ s
State weight	Q	I_3
Timing weight	R	I_8

corresponding to binary physical modes 0 through 7. The nominal switching schedule in microseconds is

$$\bar{\tau} = (0, 66, 88, 110, 132, 154, 220, 242, 264, 286). \quad (32)$$

The current configuration defines the cycle-boundary state of the target periodic orbit as

$$\mathbf{x}^* = [9.9247, 19.2928, 0.9823]^\top. \quad (33)$$

The desired average operating point is $[10, 20, 1]^\top$; it is used to design the orbit but not as the cycle-boundary feedback target.

These values differ from the older thesis snapshot, which reports $[9.5143, 19.8211, 1.0314]^\top$ and a slightly different switching schedule [3]. To keep the numerical study reproducible, this article uses the current MATLAB configuration consistently. The result-generation script also passes 3 μ s, rather than 3×10^{-6} μ s, to the controller interface because that interface receives microseconds.

4.2 Parameters of the Simulation

The exact segment propagation in (4) is used for 80 cycles. The initial state is

$$\mathbf{x}_0 = [7.51, 20.82, 0.03]^\top. \quad (34)$$

Controlled and open-loop runs use the same initial state and nominal mode sequence. To compare states with different units and magnitudes, the reported cycle-to-cycle metric is

$$\varepsilon_k = \|\text{diag}(10, 20, 1)^{-1}(\mathbf{x}_k - \mathbf{x}^*)\|_2. \quad (35)$$

The experiment is generated by `scripts/generate_results.m`; scalar values used in the text are emitted to `results/metrics.tex`. The same script constructs $\mathcal{C}$ from (27), computes $\mathcal{C}_\infty$ by invariant-set iteration, and exports their vertices and half-space data with the local closed-loop matrix.

5 Results

Figure 1 compares the continuous state trajectories. The controlled response approaches the neighborhood of the target orbit, whereas the open-loop response remains associated with a different cycle-boundary behavior over the simulated interval.

The normalized error in figure 2 decreases from 0.9854 to 0.0086 after 80 cycles, corresponding to a 99.1% reduction. The open-loop error ends at 0.2344, a reduction of only 76.2% over the same

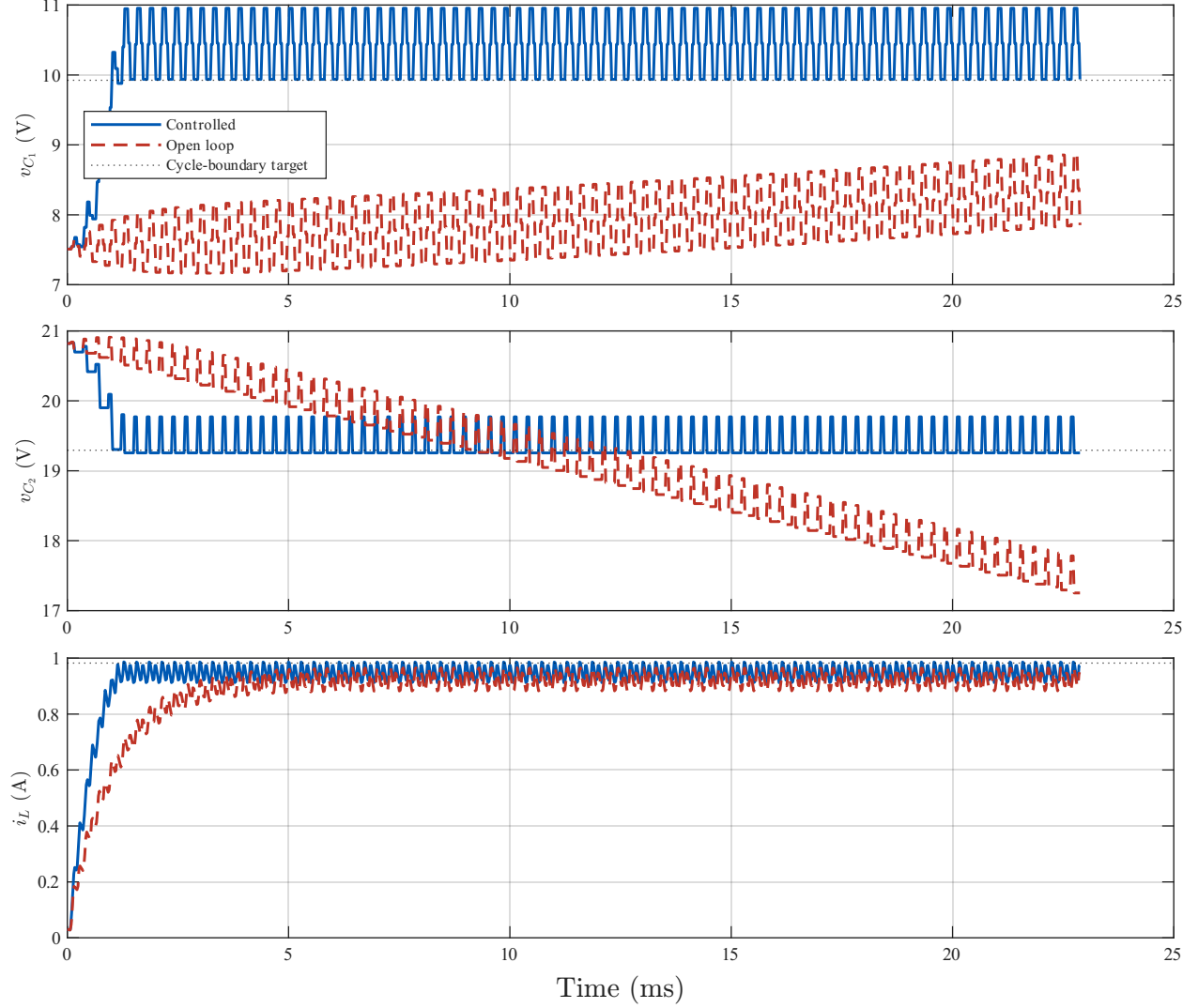


Figure 1: State response with proportional switching-time control and in open loop. The dotted values mark the cycle-boundary target, not the complete periodic waveform.

interval. The nonzero controlled residual is consistent with the nonlinear periodic plant, the local first-order model, and activation of the timing conditioner.

Figure 3 shows the manipulated variables and the shortest dwell interval in each of the first 20 cycles. The largest applied switching-time perturbation is 19.0 μ s. The shortest observed dwell interval over the complete run is 3.0 μ s, equal to the prescribed lower bound. The result numerically confirms that conditioning prevents infeasible schedules for this experiment.

For the computed gain, the local unconditioned matrix has spectral radius $\rho(A_{cl}) = 5.86 \times 10^{-8} < 1$. The invariant-set iteration returns $\mathcal{C}_\infty = \mathcal{C}$ to numerical tolerance, and the resulting polyhedron is full-dimensional in the three error coordinates. Thus, every error in the displayed volume remains in it under the local linear model, and the raw LQR action stays dwell-admissible. Figure 4 presents the region and trajectory using orthographic XY, XZ, and YZ projections together with the perspective XYZ view. The simulated initial error lies outside this set. Conditioning is active for the first 3 cycles; the trajectory enters $\mathcal{C}$ at cycle 4 and remains inside for the rest of the run.

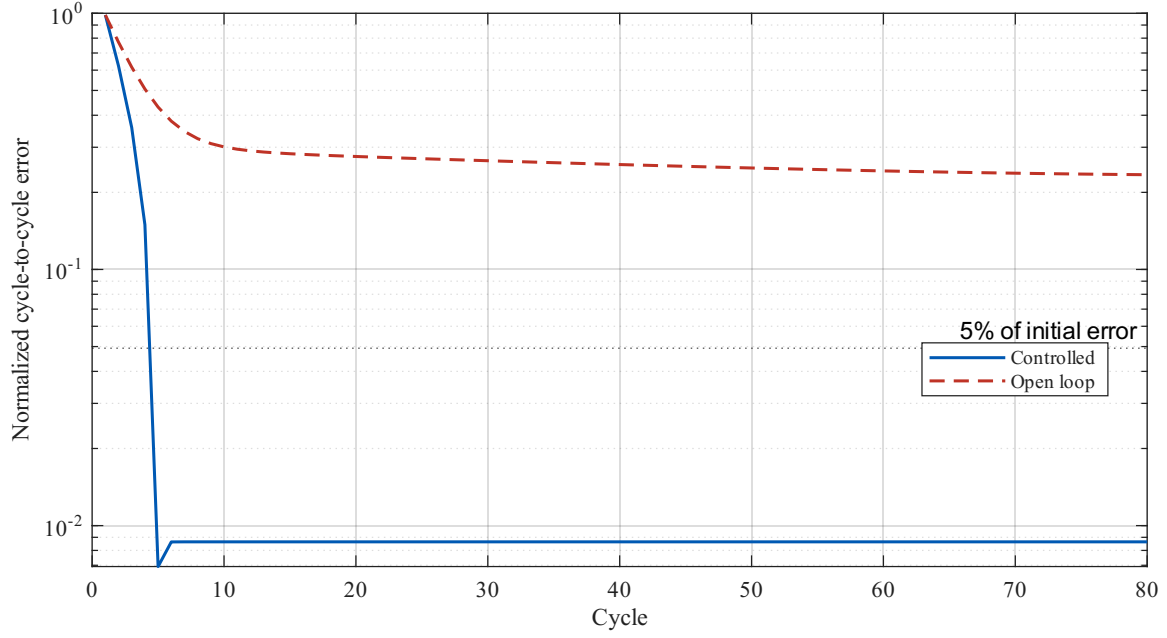


Figure 2: Normalized cycle-to-cycle error defined by (35).

The state-space view in figure 5 emphasizes that the controller does not drive the continuous state to a static point. It aligns the cycle boundary with the selected phase of the target periodic orbit while the state continues to traverse the orbit within every switching cycle.

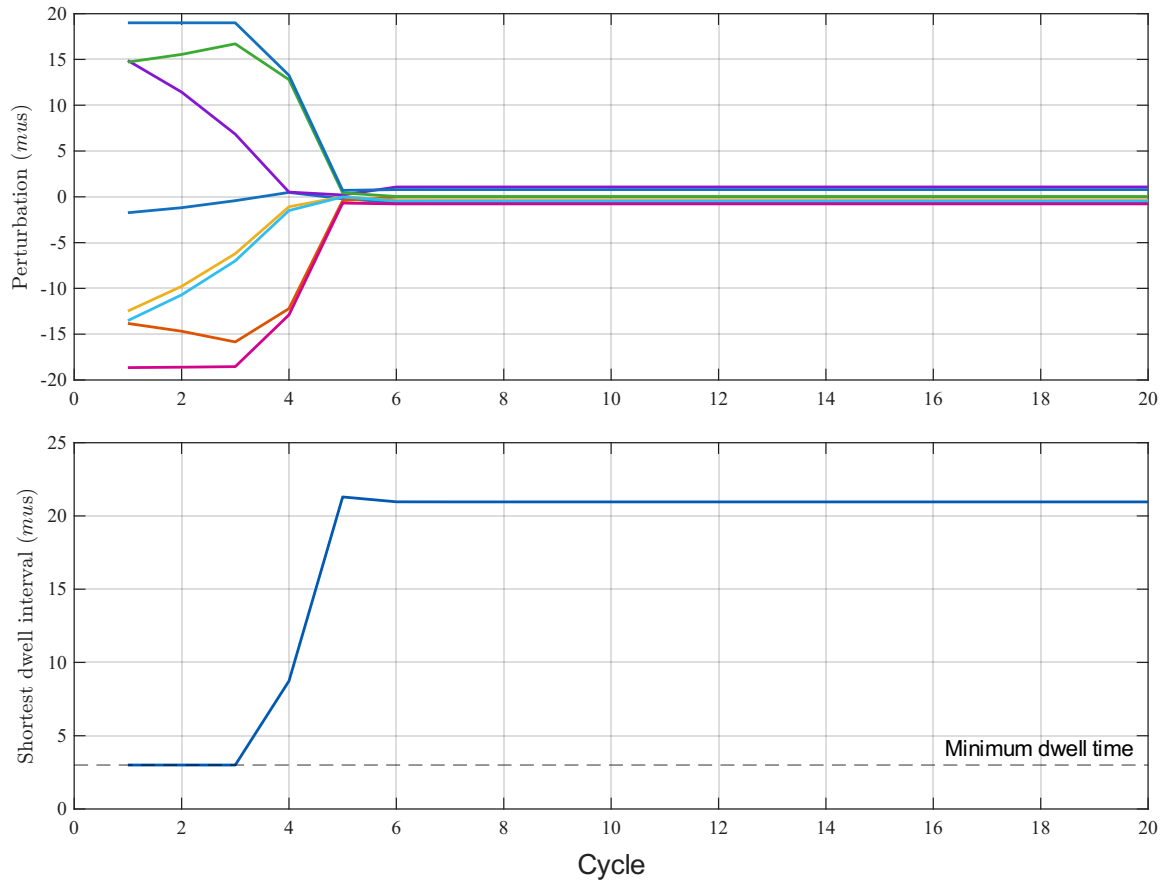


Figure 3: Applied switching-time perturbations and shortest dwell interval per cycle.

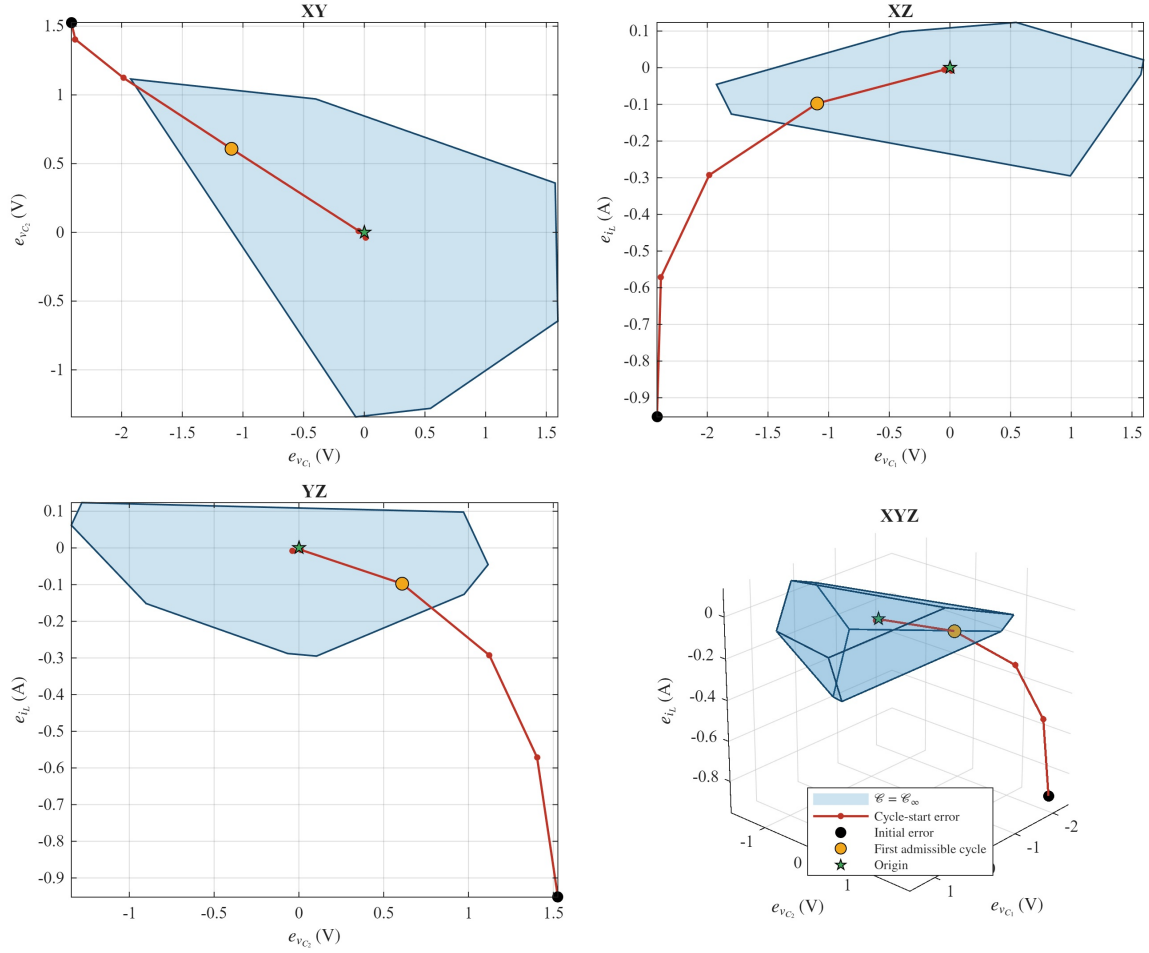


Figure 4: Orthographic and perspective views of the raw-action admissible region and the simulated cycle-start error trajectory.

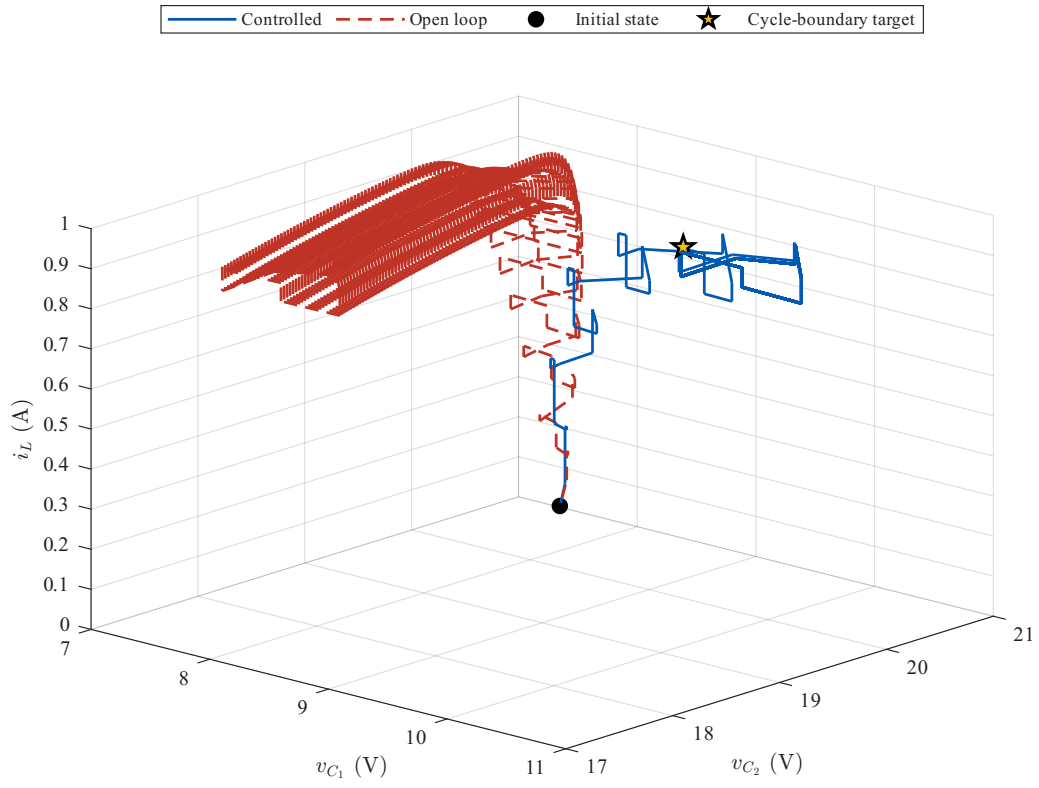


Figure 5: Controlled and open-loop trajectories in converter state space.

6 Discussion

The simulation supports the use of proportional switching-time feedback as a low-cost baseline. Once K is computed offline, the online controller does not depend on a prediction horizon or an optimization solver. It also makes the actuator constraint explicit in timing coordinates, where dwell-time feasibility can be checked directly.

Four limitations qualify the result. First, the one-cycle model is a first-order approximation around one phase of one periodic orbit, so neither $\mathcal{C}_\infty$ nor the feedback law provides a global convergence guarantee for the nonlinear plant. Second, the uniform factor in (23) is conservative: one critical dwell interval scales all perturbations. Third, conditioning changes the effective closed-loop gain and therefore invalidates the unconstrained LQR optimality claim whenever $\beta_k < 1$. Fourth, the present evidence is deterministic simulation only; measurement noise, parameter uncertainty, computation delay, timer quantization, and switching losses remain outside the experiment.

The controller should consequently be interpreted as a constrained baseline and a possible fallback law, not as a replacement for predictive control in all operating regions. A useful next comparison would hold the target orbit, model, initial conditions, and timing constraints fixed while measuring convergence, computation time, the present controller's raw-action admissible region, and the dwell-time-constrained MPC feasibility region.

7 Conclusion

A proportional controller for periodic-orbit stabilization of switched-affine systems was formulated directly in switching-time coordinates. The controller uses a discrete LQR gain on the local one-cycle error model and a scalar conditioning operation to satisfy minimum dwell time. The raw-action admissible polyhedron provides the counterpart to an MPC feasibility analysis: inside it, conditioning is inactive, and its invariant subset certifies local linear-model convergence under the unmodified gain. For the present configuration, the admissible and invariant regions coincide. In the three-cell converter simulation, the controller reduces the normalized cycle-to-cycle error by 99.1% over 80 cycles while maintaining the $3\mu\text{s}$ dwell-time bound and entering the admissible region after 3 conditioned cycles. The approach is computationally simple and physically interpretable, but remains local and can become conservative when its uniform conditioner is active. Experimental validation and a controlled comparison with constrained MPC are required before stronger robustness or real-time claims can be made.

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